

Métodos Matemáticos de la Física II

Part II: Analysis of Banach and Hilbert spaces

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(Dated: May 20, 2026)

Bla bla bla... resumen...

I. WHY BANACH AND HILBERT SPACES MATTER?

In Part I, metric spaces, normed vector spaces, Banach spaces, inner product spaces, and Hilbert spaces were introduced. However, the true importance of Banach and Hilbert spaces was intentionally postponed.

- Why are normed vector spaces not enough?
- Why are inner product spaces not enough?

The answer is essentially related to the convergence of infinite limiting processes.

In many areas of mathematics and physics, one works with:

- infinite sequences,
- infinite series,
- approximation procedures,
- Fourier expansions,
- variational methods, etc.

In such contexts, it is important that the limits produced by these processes remain inside the space itself. Banach spaces guarantee that limits associated with norm convergence remain within the space.

Hilbert spaces additionally provide a geometric structure based on orthogonality, allowing generalized Fourier expansions and orthogonal projections. This part, Part II, develops these ideas.

II. TOPOLOGICAL STRUCTURE OF METRIC SPACES

A. Open Balls

Let (X, d) be a metric space and let $x \in X$. The open ball centered at x with radius $r > 0$ is defined by

$$B_r(x) := \{y \in X : d(x, y) < r\}.$$

Open balls describe the local structure induced by the metric.

B. Open Sets

A subset $U \subseteq X$ is called open if for every $x \in U$, there exists $r > 0$ such that

$$B_r(x) \subseteq U.$$

Intuitively, every point of an open set can be slightly perturbed while remaining inside the set.

C. Closed Sets

A subset $F \subseteq X$ is called closed if its complement

$$X \setminus F$$

is open. Closed sets are important because they contain the limits of convergent sequences.

D. Limit Points

Let $A \subseteq X$. A point $x \in X$ is called a limit point of A if every open ball centered at x contains points of A different from x itself. That is,

$$\forall r > 0, \quad (B_r(x) \setminus x) \cap A \neq \emptyset.$$

E. Closure

The closure of a subset $A \subseteq X$, denoted by $\overline{A}$, is the set obtained by adding to A all of its limit points. Equivalently, $\overline{A}$ is the smallest closed subset containing A .

F. Dense Subsets

A subset $A \subseteq X$ is called dense in X if

$$\overline{A} = X.$$

This means that every point of X can be approximated arbitrarily well by points of A . Density is fundamental in approximation theory and Fourier analysis.

G. Separable Spaces

A metric space X is called separable if it contains a countable dense subset. That is, there exists a countable subset

$$D = \{x_1, x_2, \dots\} \subseteq X$$

such that

$$\overline{D} = X.$$

Intuitively, separability means that every element of the space can be approximated arbitrarily well using only countably many elements. This property is fundamental in approximation theory, Fourier analysis, and quantum mechanics. Examples of separable spaces include:

1. The n -th dimensional euclidean space $\mathbb{R}^n$. Basically, it is separable because $\mathbb{Q}$ is countable and dense in $\mathbb{R}$.
2. The space $C[a, b]$ of continuous functions with domain $[a, b] \subset \mathbb{R}$. See appendix B.
3. The space $L^2[a, b]$ of square integrable functions with domain $[a, b] \subset \mathbb{R}$.

H. Compact Sets

A subset $K \subseteq X$ is called compact if every open cover of K admits a finite subcover. In metric spaces, compactness is equivalent to sequential compactness.

I. Sequential Compactness

A subset $K \subseteq X$ is sequentially compact if every sequence in K possesses a convergent subsequence whose limit belongs to K . Compactness plays a central role in existence theorems and approximation methods.

J. Intuition behind Compactness

To develop some intuition, let us state the *Heine-Borel theorem*. It says that, for any $K \subseteq \mathbb{R}^n$,

$$K \text{ is compact} \iff K \text{ is closed and bounded.}$$

Thus:

- bounded = cannot escape to infinity,
- closed = contains its limit points.

Together:

- sequences cannot diverge,
- limits cannot leave the set.

So compactness means:

every infinite process remains controlled.

III. INFINITE SERIES AND BANACH SPACES

A. Infinite Series in Normed Spaces

Let $(V, |\cdot|)$ be a normed vector space and let $\{v_n : n \in \mathbb{N}\}$ be a sequence in V . An infinite series is an expression of the form

$$\sum_{n=1}^{\infty} v_n.$$

The series is said to converge if the sequence of partial sums

$$S_N := \sum_{n=1}^N v_n$$

converges in norm. That is, there exists $S \in V$ such that

$$\lim_{N \rightarrow \infty} |S_N - S| = 0.$$

B. The Cauchy Criterion for Series

A series converges if and only if the sequence of partial sums is a Cauchy sequence. That is,

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N} \text{ such that } m, n \geq N \Rightarrow |S_n - S_m| < \varepsilon.$$

However, a Cauchy sequence does not necessarily converge in an arbitrary normed vector space. This is precisely why completeness becomes important.

C. Why Normed Spaces Are Not Enough

In an incomplete normed vector space, a convergent approximation process may produce a limit that lies outside the space itself. Thus, one may construct:

- convergent sequences,
- convergent series,
- approximation procedures,

whose limits do not belong to the original space. This is mathematically inconvenient and physically undesirable.

D. Banach Spaces

Banach spaces solve this problem. Because Banach spaces are complete, every Cauchy sequence converges to an element of the space itself. Consequently:

- convergent infinite series remain inside the space,
- approximation procedures remain meaningful,
- limiting processes are internally consistent.

This is one of the main reasons Banach spaces are fundamental in analysis.

E. Closed Subspaces of Banach Spaces

Let V be a Banach space and let $W \subseteq V$ be a closed vector subspace. Then W is itself a Banach space under the induced norm. The closedness condition is essential because limits of convergent sequences in W must remain inside W .

IV. ORTHOGONALITY AND HILBERT SPACES

A. Orthogonality

Let $(H, \langle \cdot, \cdot \rangle)$ be an inner product space. Two vectors $u, v \in H$ are said to be orthogonal if

$$\langle u, v \rangle = 0.$$

Orthogonality generalizes the notion of perpendicularity from Euclidean geometry.

B. Orthonormal Sets

A subset $\{e_i : i \in I\} \subseteq H$ is called orthonormal if

$$\langle e_i, e_j \rangle = \delta_{ij}.$$

That is:

- each vector has norm one,
- different vectors are orthogonal.

C. Fourier Coefficients

Let $\{e_n\}_{n=1}^{\infty}$ be an orthonormal set and let $v \in H$. The numbers

$$c_n := \langle e_n, v \rangle$$

are called the Fourier coefficients of v relative to the orthonormal set. They measure the components of v along the orthogonal directions e_n .

D. Fourier Expansions

One may attempt to reconstruct v through the infinite series

$$\sum_{n=1}^{\infty} \langle e_n, v \rangle e_n.$$

This is a generalized Fourier expansion. However, such an expansion only becomes fully meaningful if the series converges within the space itself. This is why completeness is essential.

E. Bessel Inequality

Let $\{e_n\}_{n=1}^{\infty}$ be an orthonormal set in a Hilbert space H . Then, for every $v \in H$,

$$\sum_{n=1}^{\infty} |\langle e_n, v \rangle|^2 \leq |v|^2.$$

This inequality guarantees that the Fourier coefficients are square summable.

F. Parseval Identity

If the orthonormal set forms a complete orthonormal basis, then

$$|v|^2 = \sum_{n=1}^{\infty} |\langle e_n, v \rangle|^2.$$

This identity generalizes the Pythagorean theorem to infinite-dimensional spaces.

G. Why Inner Product Spaces Are Not Enough?

An arbitrary inner product space may fail to contain the limits of orthogonal expansions. Thus, generalized Fourier series may converge to objects lying outside the space. Hilbert spaces solve this problem because they are complete. Therefore:

- orthogonal expansions converge inside the space,
- generalized Fourier analysis becomes internally consistent,
- infinite-dimensional Euclidean geometry becomes possible.

This is one of the main reasons Hilbert spaces are fundamental in mathematics and physics.

V. ORTHOGONAL PROJECTIONS AND BEST APPROXIMATION

A. Orthogonal Complement

Let $M \subseteq H$ be a subset of a Hilbert space. The orthogonal complement of M is defined by

$$M^\perp := \{v \in H : \langle v, m \rangle = 0 \text{ for all } m \in M\}.$$

B. Projection Theorem

Let M be a closed subspace of a Hilbert space H . Then every vector $v \in H$ can be uniquely decomposed as

$$v = m + w,$$

where

- $m \in M$,
- $w \in M^\perp$.

Equivalently,

$$H = M \oplus M^\perp.$$

C. Best Approximation Theorem

Let M be a closed subspace of a Hilbert space H . For every $v \in H$, there exists a unique vector $m \in M$ such that

$$|v - m| = \min_{u \in M} |v - u|.$$

Thus, orthogonal projections provide optimal approximations.

D. Importance of Completeness

The projection theorem fundamentally depends on completeness. Without completeness:

- minimizing sequences may fail to converge,
- orthogonal projections may fail to exist,
- approximation procedures may break down.

Hilbert spaces guarantee that these geometric constructions remain meaningful.

VI. LINEAR OPERATORS

A. Linear Operators

Let V and W be vector spaces. A function

$$T : V \rightarrow W$$

is called linear if

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$$

for all $u, v \in V$ and all scalars α, β .

B. Bounded Operators

Let V and W be normed vector spaces. A linear operator $T : V \rightarrow W$ is called bounded if there exists $C > 0$ such that

$$|T(v)| \leq C|v|$$

for every $v \in V$.

C. Continuity of Linear Operators

A linear operator between normed vector spaces is continuous if and only if it is bounded. Thus, boundedness is the natural notion of regularity for linear operators in functional analysis.

D. Operator Norm

The operator norm of a bounded linear operator T is defined by

$$|T| := \sup_{|v| \leq 1} |T(v)|.$$

The operator norm measures the maximal amplification produced by the operator.

E. Dual Spaces and the Riesz Representation Theorem

Let V be a normed vector space. A bounded linear functional on V is a bounded linear operator

$$\varphi : V \rightarrow \mathbb{R}$$

(or $\mathbb{C}$). The set of all bounded linear functionals on V is called the dual space of V , and is denoted by V^* . The dual space plays a central role in functional analysis because it describes how vectors may be probed through linear measurements.

F. The Riesz Representation Theorem

One of the most remarkable properties of Hilbert spaces is that every bounded linear functional may be represented through the inner product. More precisely, let H be a Hilbert space and let

$$\varphi \in H^*.$$

Then there exists a unique vector $u \in H$ such that

$$\varphi(v) = u \cdot v$$

for every $v \in H$. Moreover,

$$|\varphi| = |u|.$$

Thus every bounded linear functional on a Hilbert space is obtained by taking an inner product with a fixed vector.

G. Musical Isomorphism

The Riesz theorem defines a natural correspondence

$$H \longrightarrow H^*, \quad u \mapsto u^\flat,$$

where

$$u^\flat(v) := u \cdot v.$$

This map is called the musical isomorphism. Conversely, every functional $\varphi \in H^*$ uniquely determines a vector

$$\varphi^\sharp \in H$$

such that

$$\varphi(v) = \varphi^\sharp \cdot v.$$

Thus the inner product establishes a natural identification between vectors and linear functionals. Moreover, the musical isomorphism induces an inner product over H^* . Namely, if φ and ϕ are functionals in H^* , then

$$\varphi \cdot \phi := \varphi^\sharp \cdot \phi^\sharp$$

defines the induced inner product over H^* .

Example: For $H = \mathbb{C}^n$,

$$\varphi(v) = a^\dagger v = \sum_{i=1}^n a_i^* v_i$$

for some unique vector $a \in \mathbb{C}^n$. The Riesz theorem says that Hilbert spaces behave exactly like this, even in infinite dimensions.

H. Geometric Interpretation

The Riesz theorem shows that, in Hilbert spaces, linear measurements are completely determined by geometry. Indeed, evaluating a functional amounts to projecting onto a fixed direction through the inner product. This is one of the deepest reasons why Hilbert spaces possess such a rich structure compared to general Banach spaces.

I. Final Remarks

Banach and Hilbert spaces provide the natural setting for infinite-dimensional analysis. Banach spaces guarantee the stability of limiting processes under norm convergence. Hilbert spaces additionally provide a geometric structure based on orthogonality, projections, and generalized Fourier expansions. These structures form the mathematical foundation of large areas of modern mathematics and theoretical physics.

Appendix A: $C[a, b]$ is a Banach space

Let us show that $C[a, b]$ is a Banach space under the *uniform norm*.

$$|f|_\infty := \sup_{x \in [a, b]} |f(x)|.$$

For this, we should show that $C[a, b]$ is complete under such norm. In other words, we should show that every Cauchy sequence in $C[a, b]$ converges to a function in $C[a, b]$.

That $\{f_n : n \in \mathbb{N}\}$ is a Cauchy sequence under the uniform norm means that, for any $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$\|f_n - f_m\|_\infty < \epsilon$$

for every $n, m \geq N$. This is equivalent to say that

$$|f_n(x) - f_m(x)| < \epsilon$$

holds for every $x \in [a, b]$ and every $n, m \geq N$. As a consequence, for any $x \in [a, b]$, the sequence $\{f_n(x) : n \in \mathbb{N}\}$ is Cauchy and, therefore, it is also convergent because $\mathbb{R}$ is complete. Let $f(x)$ denote the value to which $\{f_n(x) : n \in \mathbb{N}\}$ converges. Now, it remains to show that:

1. the sequence $\{f_n(x) : n \in \mathbb{N}\}$ converges to f , and
2. that the function $f : [a, b] \ni x \rightarrow f(x) \in \mathbb{R}$ belongs to $C[a, b]$ or, in other words, that f is continuous in $[a, b]$.

The condition 1. follows easily because the fact that $|f_n(x) - f_m(x)| < \epsilon$ for each $n, m \geq N$ and $x \in [a, b]$ is equivalent to say that $\|f_n - f_m\|_\infty < \epsilon$ for each $n, m \geq N$.

The condition 2. follows by taking $m \rightarrow \infty$, so that $f_m(x) \rightarrow f(x)$ and, therefore, we get that

$$|f_n(x) - f(x)| < \epsilon$$

for all $n \geq N$ and $x \in [a, b]$. Finally, this is equivalent to say that

$$\|f_n - f\|_\infty < \epsilon$$

for every $n \geq N$ and, crucially, by a fundamental theorem of calculus, this implies that f is continuous because each of the f_n is continuous by assumption.

Appendix B: $C[a, b]$ is separable

This is one of the most important examples of a separable Banach space. The key idea is that every continuous function can be uniformly approximated by polynomials with rational coefficients. Since there are only countably many such polynomials, they form a countable dense subset of $C[a, b]$. The proof relies on the Weierstrass approximation theorem.

Theorem: The space $C[a, b]$ equipped with the supremum norm

$$\|f\|_\infty = \sup_{x \in [a, b]} |f(x)|$$

is separable.

Proof:

Step 1 — A Countable Family of Polynomials. Consider the set $\mathcal{P}_\mathbb{Q}$ of all polynomials with rational coefficients:

$$p(x) = a_0 + a_1x + \cdots + a_nx^n, \quad a_k \in \mathbb{Q}.$$

We first show that this set is countable. For each degree n , the set of such polynomials is: $\mathbb{Q}^{n+1}$. Since:

- $\mathbb{Q}$ is countable,
- finite products of countable sets are countable,

it follows that: $\mathbb{Q}^{n+1}$ is countable. Now:

$$\mathcal{P}_\mathbb{Q} = \bigcup_{n=0}^{\infty} \mathbb{Q}^{n+1}.$$

A countable union of countable sets is countable. Therefore: $\mathcal{P}_\mathbb{Q}$ is countable.

Step 2 — Density of Rational Polynomials: We now show that:

$$\overline{\mathcal{P}_{\mathbb{Q}}} = C[a, b].$$

Let $f \in C[a, b]$ and let $\varepsilon > 0$. By the Weierstrass approximation theorem, there exists a polynomial

$$p(x) = a_0 + a_1x + \cdots + a_nx^n$$

with real coefficients such that:

$$|f - p|_{\infty} < \frac{\varepsilon}{2}.$$

Now approximate each real coefficient a_k by a rational number q_k sufficiently accurately. Define:

$$q(x) = q_0 + q_1x + \cdots + q_nx^n.$$

Because polynomial evaluation depends continuously on the coefficients, the approximation can be chosen so that:

$$|p - q|_{\infty} < \frac{\varepsilon}{2}.$$

Then:

$$|f - q|_{\infty} \leq |f - p|_{\infty} + |p - q|_{\infty} < \varepsilon.$$

Since $q \in \mathcal{P}_{\mathbb{Q}}$, we conclude that $\mathcal{P}_{\mathbb{Q}}$ is dense in $C[a, b]$.

Conclusion: The space $C[a, b]$ contains a countable dense subset. Therefore: $C[a, b]$ is separable.

Intuition: This theorem says something remarkable:

every continuous function can be approximated arbitrarily well using only countably many functions.

Even though:

- $C[a, b]$ is infinite-dimensional,
- and contains uncountably many functions,

its topology is still controlled by a countable set of approximating functions. This is one of the reasons Fourier analysis and numerical approximation become possible.

Appendix C: $L^2[a, b]$ is complete

The completeness of $L^2[a, b]$ is one of the fundamental results of functional analysis. However, the proof is subtler than for $C[a, b]$, because convergence in L^2 is not pointwise convergence and functions in L^2 are equivalence classes modulo sets of measure zero. Still, the core idea is the same:

a Cauchy sequence should converge to an element still inside the space.

Let us prove it carefully while keeping the intuition visible.

1. The Space $L^2[a, b]$

Recall that:

$$L^2[a, b] = \left\{ f : [a, b] \rightarrow \mathbb{R} : \int_a^b |f(x)|^2 dx < \infty \right\},$$

where, by definition, **functions differing only on sets of measure zero are identified**. The norm is:

$$\|f\|_2 = \left(\int_a^b |f(x)|^2 dx \right)^{1/2}.$$

2. What Must Be Proven?

We must show:

every Cauchy sequence in $L^2[a, b]$ converges in L^2 to another element of $L^2[a, b]$.

So let, $\{f_n : n \in \mathbb{N}\}$ be Cauchy in the L^2 -norm. That is

$$\forall \varepsilon > 0, \quad \exists N \text{ such that } m, n \geq N \Rightarrow |f_n - f_m|_2 < \varepsilon.$$

Equivalently:

$$\int_a^b |f_n - f_m|^2 dx \rightarrow 0.$$

3. Extract a Rapidly Convergent Subsequence

Since $\{f_n : n \in \mathbb{N}\}$ is Cauchy, we can choose a subsequence $\{f_{n_k} : k \in \mathbb{N}\}$ such that:

$$|f_{n_{k+1}} - f_{n_k}|_2 < 2^{-k}.$$

This is a standard “rapid convergence” trick. Define:

$$g_k = f_{n_{k+1}} - f_{n_k}.$$

Then:

$$|g_k|_2 < 2^{-k}.$$

Hence:

$$\sum_{k=1}^{\infty} |g_k|_2 < \infty.$$

4. Build the Candidate Limit

Consider the series:

$$\sum_{k=1}^{\infty} g_k.$$

Formally:

$$f_{n_m} = f_{n_1} + \sum_{k=1}^{m-1} g_k.$$

The idea is:

- if the series converges,
- then the subsequence converges.

Now define:

$$G(x) = \sum_{k=1}^{\infty} |g_k(x)|.$$

Using standard measure-theoretic arguments (essentially Minkowski/Fatou), one proves:

$$G(x) < \infty$$

for almost every x . Thus:

$$\sum_{k=1}^{\infty} g_k(x)$$

converges almost everywhere. Define:

$$f(x) = f_{n_1}(x) + \sum_{k=1}^{\infty} g_k(x).$$

Then:

$$f_{n_k}(x) \rightarrow f(x)$$

almost everywhere.

5. Convergence in L^2

Now observe:

$$f - f_{n_k} = \sum_{j=k}^{\infty} g_j.$$

Using the triangle inequality:

$$|f - f_{n_k}|_2 \leq \sum_{j=k}^{\infty} |g_j|_2.$$

But:

$$\sum_{j=k}^{\infty} 2^{-j} \rightarrow 0.$$

Therefore:

$$|f - f_{n_k}|_2 \rightarrow 0.$$

So the subsequence converges in L^2 .

6. Finish the Proof

Since the original sequence $\{f_n : n \in \mathbb{N}\}$ is Cauchy, and a subsequence converges to f , the whole sequence converges to f in L^2 . Finally:

$$|f|_2 < \infty,$$

so:

$$f \in L^2[a, b].$$

Thus $L^2[a, b]$ is complete. Hence it is a Hilbert space.

7. The Key Difference from $C[a, b]$

For $C[a, b]$:

- convergence was uniform,
- limits preserved continuity.

For $L^2[a, b]$:

- convergence is integral-based,
- functions may oscillate wildly pointwise,
- only the average square error matters.

Thus L^2 -convergence is much weaker than uniform convergence.

8. Geometric Intuition

The quantity:

$$\|f - g\|_2^2 = \int_a^b |f - g|^2$$

measures the “mean square distance” between functions. A Cauchy sequence in L^2 means:

The functions become arbitrarily close on average.

Completeness guarantees that this approximation process converges to another square-integrable function. This is exactly why Hilbert spaces are the natural setting for:

- Fourier series,
- quantum mechanics,
- PDEs,
- signal processing,
- variational methods.

Appendix D: Separability of $L^2[a, b]$

The proof is conceptually similar to the separability of $C[a, b]$, but now the approximating objects are not polynomials alone. Instead, one uses:

1. step functions,
2. rational coefficients,
3. rational interval endpoints.

The key idea is:

Every L^2 -function can be approximated arbitrarily well by simple functions built from countably many parameters.

Theorem: The Hilbert space $L^2[a, b]$ is separable.

Proof:

Step 1 — A Countable Family of Step Functions: Consider the set $\mathcal{S}$ of step functions of the form

$$s(x) = \sum_{k=1}^n q_k \chi_{I_k}(x),$$

where:

- $q_k \in \mathbb{Q}$,
- each I_k is an interval with rational endpoints,
- χ_{I_k} denotes the characteristic function of I_k .

For example:

$$s(x) = 2\chi_{[0,1/2)}(x) - \frac{3}{5}\chi_{[1/2,1]}(x).$$

Step 2 — The Set $\mathcal{S}$ Is Countable: There are only countably many:

- rational numbers,
- rational endpoints,
- finite sequences of such data.

Therefore, there are only countably many step functions in $\mathcal{S}$. Hence, $\mathcal{S}$ is countable.

Step 3 — Density of Step Functions: We now show that $\mathcal{S}$ is dense in $L^2[a, b]$. A fundamental theorem of measure theory states:

Every square-integrable function can be approximated arbitrarily well in L^2 by step functions.

Thus, for every $f \in L^2[a, b]$ and every $\varepsilon > 0$, there exists a step function

$$s(x) = \sum_{k=1}^n a_k \chi_{J_k}(x)$$

such that

$$\|f - s\|_2 < \frac{\varepsilon}{2}.$$

Here:

- the coefficients a_k are real,
- the interval endpoints may be irrational.

Step 4 — Approximate the Coefficients and Intervals Rationally: Now approximate:

- each coefficient a_k by a rational number q_k ,
- each interval endpoint by rational endpoints.

This produces a rational step function $r \in \mathcal{S}$. Because the L^2 -norm depends continuously on:

- the coefficients,
- and the interval lengths,

the approximations can be chosen sufficiently accurately so that

$$|s - r|_2 < \frac{\varepsilon}{2}.$$

Then, by the triangle inequality,

$$|f - r|_2 \leq |f - s|_2 + |s - r|_2 < \varepsilon.$$

Thus:

$$\mathcal{S}$$

is dense in $L^2[a, b]$.

Conclusion: We have constructed a:

- countable,
- dense

subset of $L^2[a, b]$. Therefore, $L^2[a, b]$ is separable.

Intuition: This theorem says:

Even though $L^2[a, b]$ contains enormously complicated objects, every element can still be approximated using only countably many simple building blocks.

This is one of the reasons:

- Fourier series,
- numerical approximations,
- spectral decompositions,
- quantum mechanics,

become mathematically manageable.

Connection with Hilbert Spaces: Separability is particularly important for Hilbert spaces because it allows the existence of countable orthonormal bases:

$$f = \sum_{n=1}^{\infty} c_n e_n.$$

Without separability, one would generally need uncountably many basis vectors, and generalized Fourier analysis would become much more complicated. This is one of the reasons most Hilbert spaces appearing in physics are separable.

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